

1. Dataset provenance and description

The data employed in this project is the Yellow Taxi Trip Records dataset sourced from NYC Open data platform published by the New York City Taxi and Limousine Commission (TLC), which is publicly available at: <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>. This dataset contains large-scale real-world transactional records of taxi trips across New York City spanning over a decade (2009 - 2026). It is provided as open government data with no access restrictions or license limitations. Therefore, it is freely usable for academic work and is widely cited in existing research, industry analysis and public policy studies (Xie et al., 2021).

2. Dataset size and characteristics

The table below summarizes the characteristics and sizes of the dataset. The raw dataset was capped from Jan 2009 to December 2025 with each record representing one completed trip, which contains the information of pickup and dropoff time stamp, distance travelled, fare, payment type, vendor ID and driver-reported passenger counts.

Dataset level	Size	Format	Records / fields	Period covered	Last update
Raw	29.66 GB	Parquet	19 fields per record	Jan 2009 – Dec 2025	Ongoing, TLC publishes monthly with 2 months delay to allow full vendor submissions
Working	12.93 GB	Parquet	144 files, 19 fields	Jan 2014 – Dec 2025	Dec 2025
PySpark processing	4.96 GB	Parquet	143 files (1 failed to download), 19 fields; filtered to night-time trips only (8pm–5am)	Jan 2014 – Dec 2025	Dec 2025

3. Data preparation and quality

Before analysis, a series of data quality and preparation checks were performed on the dataset. More details can be found in the assignment's Appendix and GitHub repository. Following the data quality principles outlined by Sharifnia et al. (2025), several checks were completed to improve data completeness, consistency, accuracy and reliability.

Column header consistency was reviewed and a difference between `Airport_fee` and `airport_fee` was identified and standardised. **Missing values** for `airport_fee` and `congestion_surcharge` were treated as indicating that no charge applied, while `RatecodeID` was viewed as an "unknown" category. Records with missing `passenger_count` values were retained but may need to be filtered for analyses involving this variable.

Dataset size over time was measured to assess **completeness** and **trends**. A significant decline in records was observed during 2020, aligning with the COVID-19 pandemic. Duplicate records were also assessed; however, the dataset does not contain a unique identifier per driver, vehicle or passenger, therefore no records were removed.

Noisy data detection identified inconsistencies in the pickup and drop-off timestamp sequence. A small proportion of records were affected and pickup times were adjusted to 16 minutes before drop-off time, representing the average trip duration. **Outlier detection** was performed using the Z-score approach with a threshold of $|Z| > 3$. The most significant outliers were observed for `tolls_amount` and `airport_fee`, but further investigation suggested these represented genuine operational variation. Passenger counts greater than 6 were capped at 6 and values of 0 were replaced with 1.

Zero-value checks showed the highest proportions for `tolls_amount`, `tip_amount` and `airport_fee`, but these were considered normal within taxi operations. Zero values were also observed for `improvement_surcharge`, `extra`, `congestion_surcharge` and `mta_tax`, where such charges may not have been applicable.

Data type and format consistency were assessed and standardised where required. Data quality checks were performed before and after standardisation to ensure no data quality was lost.

Pickup and drop-off **geo-location distributions and trends** were examined to assess geographical coverage and consistency over time (Kostakis & Kargas, 2021). Some shifts were observed, but no major anomalies were identified.

During preprocessing to limit the dataset to night-time trips, the `yellow_tripdata_2021-10` file failed to process and was therefore excluded from the assignment.

4. Exploratory data analysis (EDA)

To identify patterns, relationships, trends and anomalies within the dataset, statistical summaries and visualisations were produced (see Appendix section 3, Exploratory data analysis- Figures 1 to 7 and Tables A1 to A4).

Trends: The average monthly volume was approximately 2.08 million trips, with a maximum of 5,256,647 (March 2014) and a minimum of 32,858 (April 2020). The data shows that the volume declined steadily from approximately 4.5 million trips per month in 2014 to around 2

million by early 2020 (Figure 1). This trend was interrupted by a sharp collapse during the COVID-19 lockdown after which volume recovered only partially, suggesting that the pandemic accelerated a longer-term decline likely driven by competition from ride-hailing services. Average fares increased from around \$12 in 2014 to between \$15 and \$22 by 2025 (Figure 2), spiking sharply in mid-2020, which is best explained by reduced trip volume relative to fixed minimum charges rather than genuine price inflation.

Anomalies: The distributional analysis revealed notable data quality concerns (Figure 4; Table A2). Although most fares and distances fall within realistic ranges, the dataset contains fares as low as -\$800 and distances of up to 12 million miles. These values are physically implausible and most likely caused by meter or GPS errors.

Relationships: Trip distance and fare show a strong positive correlation (Figure 5). This was expected given that fares are calculated primarily on a distance basis. However, distinct horizontal bands are visible at fixed fare values, most notably around \$52, which likely correspond to flat-rate pricing such as the JFK airport fare. This indicates that although distance is the dominant factor for fare prices, certain routes follow fixed pricing rules rather than the standard metered rate. This means that the relationship is not purely linear across the dataset.

Patterns: Demand peaks between 8pm and 10pm before tapering steadily into the early morning hours (Figure 6), consistent with typical night-time travel behaviour. Credit card payments dominate over cash payments. The two vendors, Curb Mobility and CMT, together account for over 99.8 percent of trips with a negligible share carrying undocumented VendorID codes (Figure 7; Table A4).

5. 7 Vs of Big Data: Analyze and discuss

The Yellow Taxi dataset is a good example of big data due to its large size, frequent generation, and varied structure. Kostakis and Kargas (2021) identify five key characteristics of big data: volume, velocity, variety, veracity, and value. These characteristics are applied to the table in the table below.

V	Dataset-specific evidence	Why it matters / relevance
Volume	Millions of records are created on a trip-level across multiple time periods. Each row represents a single taxi trip with data fields available that contains the details of the trip.	There are valuable insights to be gained from analysing this data source, for example travel demand patterns, location-based trends, etc. This is a good example of a storage, memory and processing challenge in Big Data (Kostakis & Kargas, 2021).
Velocity	With many taxis in the fleet, there is data generated continuously as trips occur. This will specially create high frequency records.	Taxi operations require quick reaction to changes. Real-time processing is crucial in enabling support towards fleet allocation, congestion monitoring etc (Kostakis & Kargas, 2021).
Variety	There are different data types contained within this dataset. With data types ranging from datetime fields, numeric values and categorical features.	Various angles are available to analyse the data by when looking at the variety of data types, such as time patterns, payment behavior, etc. Data type handling is vital when processing the data (Kostakis & Kargas, 2021).
Value	This dataset can add immense value through insights such as peak travel times, average fare and tip patterns, passenger behavior across different payment types, etc.	There can be practical value derived from these insights through improved decision-making, operational efficiency and data-driven planning (Saleem & Siddik, 2025; Kostakis & Kargas, 2021).
Veracity	There might be data quality issues within the dataset, such as missing data (eg.passenger_count), zero or negative trip distances, duration, invalid timestamps, very short trips, etc.	Incorrect/incomplete data can distort analysis, showing the importance of data quality (Kostakis & Kargas, 2021). Eg, low-distance trips with high fares could mislead conclusions about pricing.
Variability	The nature of the taxi operations (demand and trip characteristics) will result in variability by time of day, season, day of the week, trip purpose, etc.	The avoidance of over-generalisation is important as data behaviour may vary across time periods and operational conditions (Kostakis & Kargas, 2021). For example, merely using averages will hide important insights.
Visualisation	There are endless visualisations possible with this dataset, such as heatmaps of zones, line charts of trips by hour, bar charts of payment types, boxplots of fare amounts, etc.	The location, fare and timestamp variables can support heatmaps, hourly demand charts, fare boxplots, etc. Visualisation converts the large data into understandable patterns to interpret (Saleem & Siddik, 2025).

6. Business or societal value and insights

Business value: This dataset has the potential to enable business on travel demand, passenger behavior, revenue patterns, pricing, etc. These insights can support data-driven decision-making and enable organisations to improve operational efficiency and strategic planning (Kostakis & Kargas, 2021).

Societal value: There is potential to understand urban mobility and enable transport planning. Environmental initiatives can be supported by identifying traffic congestion hot spots to point to the implementation of alternative transport solutions (Kostakis & Kargas, 2021).

Limitations: The Yellow Taxi dataset has several limitations. Data quality (missing values, incorrect timestamps, etc) require careful validation and preprocessing (Sharifnia et al., 2025). Representability, like only considering one transport provider (like Yellow Taxi) in a big city will only give partial insights as there are more providers and modes that exist (Kostakis & Kargas, 2021). Without the consideration of external factors (like weather, special events, road incidents, etc) there might be lack of context towards certain trends.

References

- Kostakis, P., &Kargas, A. (2021). Big-data management: A driver for digital transformation? *Information*, 12(10), 411. <https://doi.org/10.3390/info12100411>
- Saleem, P. M. B., &Siddik, M. M. (2025). *Big Data Management: Principles, Practices and Future*. In *Recent Trends in Multidisciplinary Research (RTMR-2025)* (pp. 281-290). Association of Global Academician and Researchers (AGAR) Publications. ISBN 978-93-7013-138-5.
- Sharifnia, A. M., Kpormegbey, D. E., Thapa, D. K., & Cleary, M. (2025). *A primer of data cleaning in quantitative research: Handling missing values and outliers*. *Journal of Advanced Nursing*, 82(1), 970-975. <https://doi.org/10.1111/jan.16908>
- Xie, C., Yu, D., Zheng, X., Wang, Z., & Jiang, Z. (2021). Revealing spatiotemporal travel demand and community structure characteristics with taxi trip data: A case study of New York City. *PLoS one*, 16(11), e0259694.

Appendix

1. Data Preparation and Quality

i. Data Familiarisation

Before the preprocessing was applied to the dataset, initial checks were performed to understand the proportions of certain variables by looking at one file, 2014-01. See below trips per Rate Code, Passenger Count, Payment Type and Day vs Night trips:

Rate Code:	Passenger Count:	Payment Type:	Day vs Night time trips:
Rate Code Distribution for 2014-01 rate_label Standard Rate13490929 JFK227834 Negotiated Fare38130 Newark19380 Nassau/Westchester4550 Group Ride145	Passenger Count Distribution for 2014-01 passenger_count 0259 19727321 21891588 3566248 4267540 5789070 6540444 77 85 916 20819	Payment Type Distribution for 2014-01 payment_label Credit Card7855148 Cash5818066 Unknown67880 No Charge32076 Dispute9347	Day vs. Night Trip Counts for 2014-01 time_of_day Day7364425 Night6418092

ii. Data Preparation & Quality

Below are table outputs from the data quality checks performed on the dataset, these elements are discussed in detail in Question 3 of the assignment.

Missing values:

	TotalMissingCount	NumFilesAffected	AvgMissingPercentage
Column			
airport_fee	251643699	143	65.424731
congestion_surcharge	222611697	143	48.073465
store_and_fwd_flag	24606559	96	12.133227
improvement_surcharge	18432082	5	79.999036
RatecodeID	7550472	89	9.106397
passenger_count	7550472	89	9.106397

Duplicate records:

Top 10 Files with Highest Percentage of Duplicates:			
File	Total Records	Number of Duplicates	Percentage of Duplicates
18 yellow_tripdata_2020-01.parquet	1737021	1763	0.101496
17 yellow_tripdata_2018-10.parquet	2594690	144	0.005550
19 yellow_tripdata_2020-06.parquet	91009	1	0.001099
3 yellow_tripdata_2014-04.parquet	4947535	47	0.000950
6 yellow_tripdata_2014-07.parquet	4483303	35	0.000781
2 yellow_tripdata_2014-03.parquet	5256647	41	0.000780
4 yellow_tripdata_2014-05.parquet	5151841	39	0.000757
0 yellow_tripdata_2014-01.parquet	4550300	34	0.000747
1 yellow_tripdata_2014-02.parquet	4488391	29	0.000646
5 yellow_tripdata_2014-06.parquet	4821908	26	0.000539

Pick up and drop off time inconsistencies:

Top 10 Files with Highest Percentage of Inconsistent Records:			
File	Nr Inconsistent Records	Percentage of Inconsistent Records	
63 yellow_tripdata_2021-01.parquet	601	0.358290	
64 yellow_tripdata_2021-02.parquet	645	0.342981	
61 yellow_tripdata_2020-11.parquet	627	0.273807	
62 yellow_tripdata_2020-12.parquet	455	0.242475	
65 yellow_tripdata_2021-03.parquet	621	0.206195	
60 yellow_tripdata_2020-10.parquet	446	0.156017	
59 yellow_tripdata_2020-09.parquet	314	0.141334	
111 yellow_tripdata_2025-11.parquet	1433	0.115237	
103 yellow_tripdata_2024-11.parquet	1014	0.098144	
83 yellow_tripdata_2022-11.parquet	789	0.090094	

Outlier summary:

	TotalZscoreOutliers	NumFilesAffected	AvgPercentageZscoreOutliers
Column			
tolls_amount	12193463	143	4.581931
passenger_count	8823532	143	2.973423
tip_amount	5356374	143	1.875509
total_amount	5264392	143	1.895562
fare_amount	5047590	143	1.646346
extra	4766270	143	1.349187
trip_distance	3536773	143	0.940372
airport_fee	2548544	56	5.004044
mta_tax	2523434	143	1.318688
congestion_surcharge	2520172	83	2.146678
improvement_surcharge	1256091	137	0.723348

Zero values:

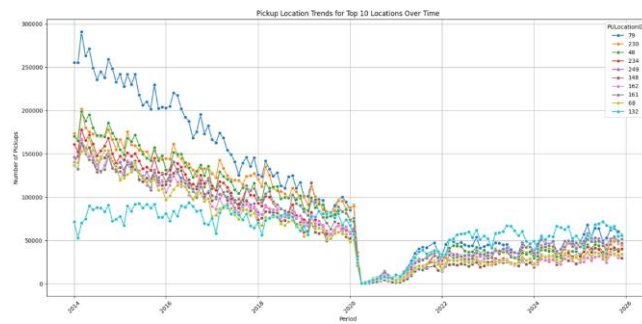
Column	TotalZeroCount	NumFilesAffected	AvgPercentageZero
tolls_amount	284687649	143	94.943050
tip_amount	103782181	143	32.894285
improvement_surcharge	38544687	141	5.789442
airport_fee	14881442	25	74.306276
extra	14809262	143	8.330038
congestion_surcharge	7020348	84	8.572921
trip_distance	2593840	143	1.233696
mta_tax	1692049	143	0.757712
passenger_count	1338501	143	0.854636
fare_amount	104779	138	0.042870
total_amount	37554	138	0.016258

Data type consistency:

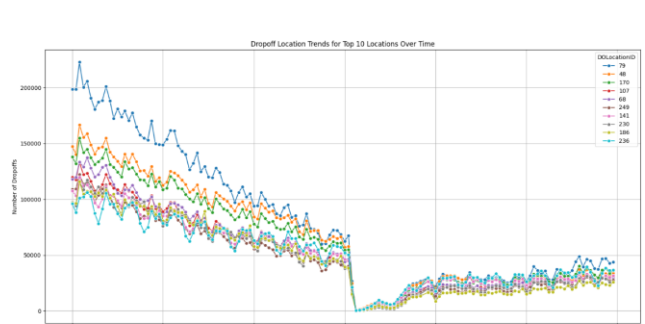
Inconsistencies for the following columns:

- Column 'VendorID': Found types ['int32', 'int64']
- Column 'passenger_count': Found types ['double', 'int64']
- Column 'RatecodeID': Found types ['double', 'int64']
- Column 'PULocationID': Found types ['int32', 'int64']
- Column 'DOLocationID': Found types ['int32', 'int64']
- Column 'improvement_surcharge': Found types ['double', 'null']
- Column 'congestion_surcharge': Found types ['double', 'null']
- Column 'airport_fee': Found types ['double', 'null']

Pick up location trend:

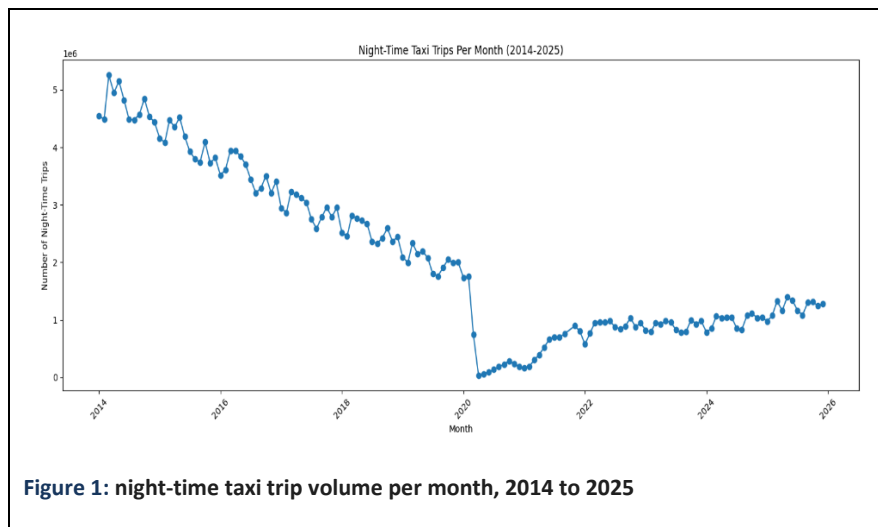


Dropoff location trend:



2. Exploratory Data Analysis (EDA):

Below are the statistical summaries and visualizations produced to analyse patterns, relationships, trends and anomalies existing in the dataset.



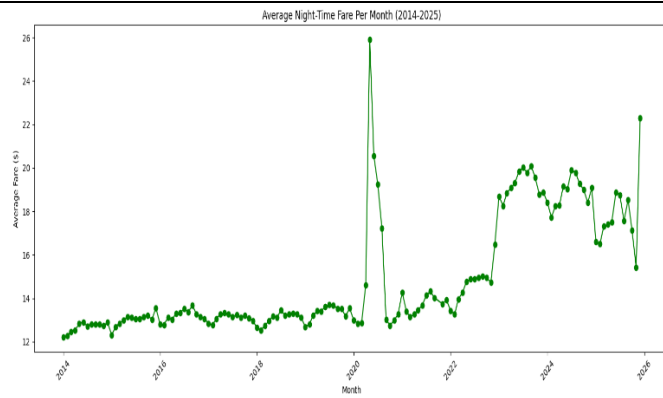


Figure 2: average night-time fare per month, 2014 to 2025

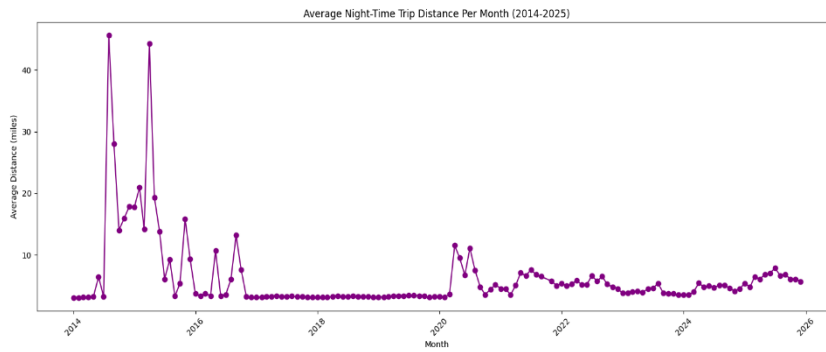


Figure 3: average night-time trip distance per month, 2014 to 2025

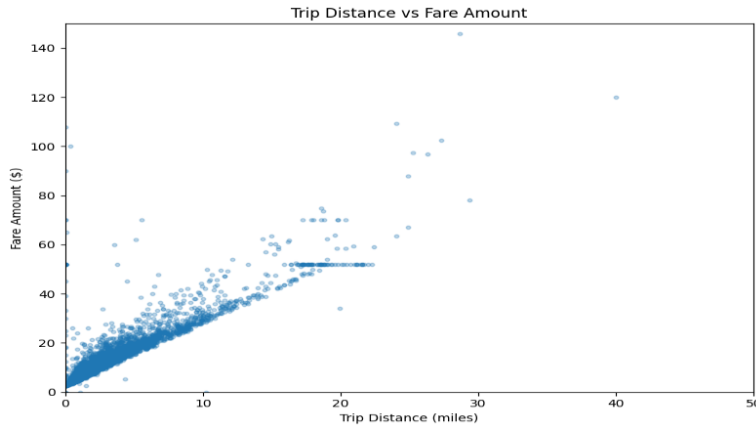


Figure 5: Scatter plot showing the relationship between trip distance and fare amount

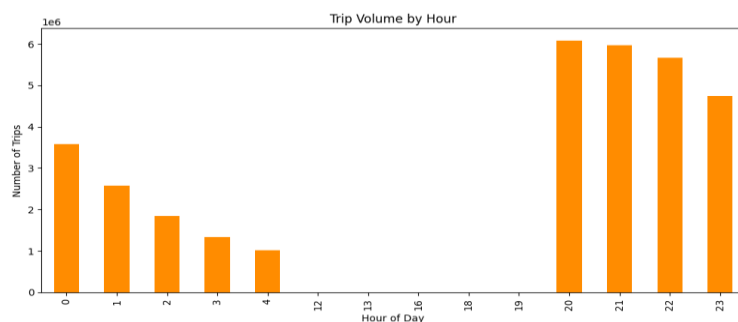
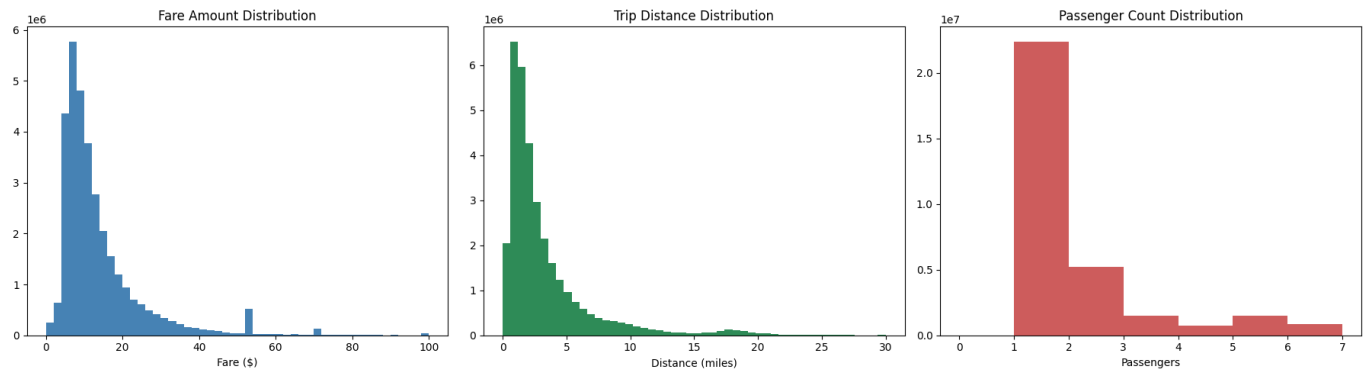
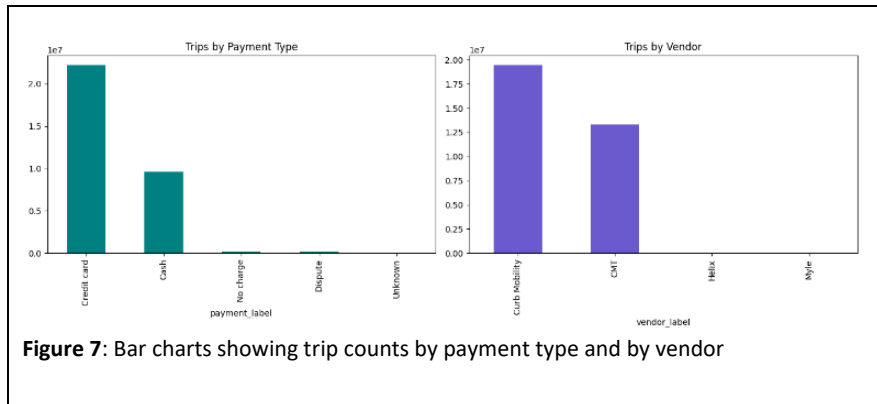


Figure 6: Bar chart of trip volume by hour of day within the night-time window



TableA1: monthly summary statistics (trip count, average fare, average distance)

Statistic	Trip Count	Average Fare (\$)	Average Distance (mi)
Count	143	143	143
Mean	2,081,185	14.88	6.21
Std Dev	1,431,983	2.70	6.10
Min	32,858	12.22	3.05
25th percentile	909,372	13.06	3.29
Median (50th)	1,760,431	13.41	4.48
75th percentile	3,206,654	17.17	6.25
Max	5,256,647	25.91	45.62

Table A2: Fare, Distance, and Passenger Count Distribution Statistics (Sampled Data)

Statistic	Fare Amount (\$)	Trip Distance (mi)	Passenger Count
Count	32,836,140	32,836,140	32,137,131
Mean	13.65	5.34	1.64
Std Dev	90.66	3.772.13	1.24
Min	-800.00	-34.94	1
25th percentile	7.00	1.14	1
Median (50th)	10.00	2.00	1
75th percentile	16.00	3.81	2
Max	505,296.80	12,000,000.00	6

Table A3: Anomaly counts (sampled data)

Anomaly	Count
Negative or zero fares	218,874
Zero-passenger trips	0
Trips over 100 miles	328
Fares over \$200	4,823

Table A4: Trip Proportion by Vendor

VendorID	Vendor	Proportion (%)
2	Curb Mobility	59.28
1	CMT	40.58
4	Undocumented	0.064
7	Helix	0.057
6	Myle	0.007
3	Undocumented	0.004
5	Undocumented	0.0001